

# Lab07\_Blumtritt

1. Baixe o arquivo uwmadison.sqlite3. Conecte-se a ele usando o pacote RSQLite, armazenando a conexão em uma variável conn.

```
library(DBI)
library(RSQLite)

conn <- dbConnect(SQLite(), "uwmadison.sqlite3")
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (subjects.abbreviation='STAT');

```
WITH profs AS (
  SELECT DISTINCT i.name FROM INSTRUCTORS I
  JOIN TEACHINGS T ON I.id = T.instructor_id
  JOIN SECTIONS SE ON SE.uuid = T.section_uuid
  JOIN SUBJECT_MEMBERSHIPS SM ON SM.course_offering_uuid = SE.course_offering_uuid
  JOIN SUBJECTS SU ON SU.code = SM.subject_code
  WHERE SU.abbreviation = 'STAT')

SELECT name, (SELECT COUNT(*) FROM profs) AS total_professores FROM profs
```

Table 1: Displaying records 1 - 10

| name            | total_professores |
|-----------------|-------------------|
| MINJING TAO     | 377               |
| DONALD PORTER   | 377               |
| SHENG WANG      | 377               |
| KUNLING HUANG   | 377               |
| DONG XIA        | 377               |
| JENNIFER NGUYEN | 377               |
| SEULKEE YUN     | 377               |

| name        | total_professores |
|-------------|-------------------|
| DUZHE WANG  | 377               |
| YONGFENG WU | 377               |
| JARED BROWN | 377               |

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

Quem é o professor mais difícil? Quem é o professor mais fácil?

```
WITH prof_dif AS (
SELECT i.name AS professor, ROUND(SUM((g.a_count * 4.0) + (g.ab_count * 3.5) + (g.b_count * 3) + (g.bc_count * 2.5) + (g.c_count * 2) + (g.d_count * 1) + (g.f_count * 0))) AS MPOND
FROM INSTRUCTORS i
JOIN SECTIONS s ON g.course_offering_uuid = s.course_offering_uuid AND g.section_number = s.section_number
JOIN TEACHINGS t ON s.uuid = t.section_uuid
JOIN INSTRUCTORS i ON t.instructor_id = i.id
JOIN SUBJECT_MEMBERSHIPS sm ON g.course_offering_uuid = sm.course_offering_uuid
JOIN SUBJECTS su ON sm.subject_code = su.code
WHERE su.abbreviation = 'STAT'
GROUP BY i.id, i.name
HAVING SUM(g.a_count + g.ab_count + g.b_count + g.bc_count + g.c_count + g.d_count + g.f_count) > 0
)
SELECT * FROM prof_dif WHERE MPOND = (SELECT MIN(MPOND) FROM prof_dif) OR MPOND = (SELECT MAX(MPOND) FROM prof_dif)
```

Table 2: 4 records

| professor                    | MPOND |
|------------------------------|-------|
| JAMES D KUELBS               | 2.638 |
| SRIKANTHMADHAVAN ARAVAMUTHAN | 4.000 |
| GUANHUA CHEN                 | 4.000 |
| YAJUAN SI                    | 4.000 |

Qual é a disciplina mais difícil? Qual é a disciplina mais fácil?

```
WITH subj_dif AS (
SELECT c.name AS disciplina, ROUND(SUM((g.a_count * 4.0) + (g.ab_count * 3.5) + (g.b_count * 3) + (g.bc_count * 2.5) + (g.c_count * 2) + (g.d_count * 1) + (g.f_count * 0))) AS MPOND
FROM COURSE_OFFERINGS co
JOIN SUBJECT_MEMBERSHIPS sm ON g.course_offering_uuid = sm.course_offering_uuid
JOIN COURSE_OFFERINGS co ON g.course_offering_uuid = co.uuid
JOIN COURSES c ON co.course_uuid = c.uuid
)
SELECT * FROM subj_dif WHERE MPOND = (SELECT MIN(MPOND) FROM subj_dif) OR MPOND = (SELECT MAX(MPOND) FROM subj_dif)
```

```

JOIN SUBJECTS su ON sm.subject_code = su.code
WHERE su.abbreviation = 'STAT'
GROUP BY c.uuid, c.name, c.number
HAVING SUM(g.a_count + g.ab_count + g.b_count + g.bc_count + g.c_count + g.d_count + g.f_coun
SELECT * FROM subj_dif WHERE MPOND = (SELECT MIN(MPOND) FROM subj_dif) OR MPOND = (SELECT MA

```

Table 3: 5 records

| disciplina                                            | MPOND |
|-------------------------------------------------------|-------|
| Nonparametric Statistics and Machine Learning Methods | 4.000 |
| Introduction to the Theory of Probability             | 2.886 |
| Sample Survey Theory and Method                       | 4.000 |
| Data Science Practicum                                | 4.000 |
| Empir Proc&Semiparmtrc Infernc                        | 4.000 |

4. Desconecte do banco de dados.

```
dbDisconnect(conn)
```

```

Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()

```

```
[1] "2026-09-20 01:24:24 -03"
```